



What is Wrong with These Experiments?



There is at least one problem with each of the following experiments. What could make them better?

1. Mr. Wilcox is trying out a new beard oil to make his beard grow faster for No Shave November. At the end of November, his beard is 3 inches longer. He concludes that since his beard grew 3 inches, the beard oil must work.
 - a. What is wrong with this experiment?
 - b. What could be done to improve this experiment?
2. The track coach decided to offer the track athletes two options for daily workouts: a rigorous strength building workout, or a relaxed workout of mostly stretching. After two months, the athletes that chose the strength building workout were 11% faster, while the relaxed workout group was only 3% faster.
 - a. What is wrong with this experiment?
 - b. Describe an appropriate method the coach could use to randomly assign the two workouts to the 120 athletes.
 - c. What is the benefit of using random assignment?
3. On a recent test day, an AP Stats teacher wanted to determine if smart pills would improve grades. As students walked into class, they were randomly assigned a pill that said either "smart pill" or "sugar pill" on it. The test scores were significantly higher for the group of students that took the smart pill.
 - a. What is wrong with this experiment?
 - b. What could be done to improve this experiment?

Lesson 2.6 – Designing Experiments

QuickNotes

Check Your Understanding

A group of researchers in Africa find a creative way to protect cattle from lion attacks – they paint eyes on the cows' rears. To determine if this treatment is effective, they randomly assign the cattle to one of three treatments: Eyes on their rear, cross-marks on their rear, or nothing on their rear. After 4 years of roaming the plains, the cows with eyes saw no deaths, the cows with a cross suffered 4 deaths, and the cows with no marks suffered 15 deaths.

- a. Explain why it was important to have a control group that didn't get a mark on their rear.
- b. Suppose the researchers had 1200 cows. Describe how to randomly assign the cows to the treatments.
- c. What is the purpose of randomly assigning treatments in this context?
- d. Create an outline showing a completely randomized design for the experiment.